

UChicago REU Notes Week 1

Zih-Yu Hsieh

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1 Week 1

Here it's mainly the preparation for any further theory in spectra. To learn the material, here I focused mainly on some topological preliminaries. Main reference comes from *Concise Course in Algebraic Topology*.

Remark 1.1. I believe most texts assume that compactness means not only the existence of finite subcover for arbitrary open cover, but also Hausdorffness on the space. Here, we also assume this.

1.1 Compactly Generated Spaces

In most literatures I've read, the spaces are assumed to be compactly generated. So, this section is collecting some information I have about it. First, starting with the notion of k -spaces:

Definition 1.2. Given a topological space X , a subset $A \subseteq X$ is k -closed, if for any compact space K , and continuous map $g : K \rightarrow X$, the preimage $g^{-1}(A) \subseteq K$ is closed.

X is a k -space, if any k -closed subset $A \subseteq X$ is closed.

Which, adapting the usual continuous maps, this realizes a *Category of k -spaces*, denoting $k\mathcal{U}$. There are some properties about this category, regarding a functor from regular category of spaces $\mathcal{U}$, and this category's completeness / cocompleteness.

Definition 1.3. Given a space X , define a topological space kX where the underlying set coincides with X , while requiring all k -closed sets $A \subseteq X$ to be closed in kX as its new topology.

This construction satisfies the closed sets axioms, hence form a topology; also, under the new topology, kX is a k -space. Furthermore, since all closed sets are automatically k -closed, the set identity function $\iota_X : kX \rightarrow X$ is in fact continuous.

We want to further argue k is a functor $\mathcal{U} \rightarrow k\mathcal{U}$, which can be achieved with the following proposition:

Proposition 1.4. *Let Z be a k -space, Y arbitrary topological space, and $f : Z \rightarrow Y$ a continuous map. Then, there exists a unique continuous map $kf : Z \rightarrow kY$, such that the following diagram commutes:*

$$\begin{array}{ccc} Z & \xrightarrow{\forall f} & Y \\ & \searrow \exists! kf & \nearrow \iota_Y \\ & kY & \end{array}$$

Proof. Since ι_Y is a set identity map, uniqueness of kf is clear (as set function it coincides with f). To show kf is continuous, it suffices to show all k -closed set $A \subseteq Y$ has preimage $f^{-1}(A) \subseteq Z$ being closed.

Let $g : K \rightarrow Z$ be arbitrary continuous map, with K being compact. Consider $f \circ g : K \rightarrow Z \rightarrow Y$, for any $A \subseteq Y$ that's k -closed, $(f \circ g)^{-1}(A) = g^{-1}(f^{-1}(A)) \subseteq K$ is closed, hence $f^{-1}(A) \subseteq Z$ is also k -closed. Since Z is a k -space, $f^{-1}(A)$ is closed. $\square$

Corollary 1.5. *Given any continuous map $f : X \rightarrow Y$, there exists a unique continuous map $kf : kX \rightarrow kY$, such that the following diagram commutes:*

$$\begin{array}{ccc} X & \xrightarrow{\forall f} & Y \\ \iota_X \uparrow & & \uparrow \iota_Y \\ kX & \xrightarrow{\exists! kf} & kY \end{array}$$

Proof. Since $f \circ \iota_X : kX \rightarrow Y$ is continuous, with kX being a k -space, **Proposition 1.4** guarantees the existence and uniqueness of kf (which as set map it also agrees with f , as ι_X is set identity). $\square$

This formulates $k : \mathcal{U} \rightarrow k\mathcal{U}$ as a functor. And, if consider the inclusion functor $\mathcal{I} : k\mathcal{U} \rightarrow \mathcal{U}$, **Proposition 1.4** also insists $(\mathcal{I}, k)$ as an adjoint pair, as it phrases a natural isomorphism as follow:

$$\text{Hom}_{\mathcal{U}}(\mathcal{I}Z, Y) \cong \text{Hom}_{k\mathcal{U}}(Z, kY), \quad f \mapsto kf$$

Using some categorical nonsense, k as a right adjoint preserves all limits, so $k\mathcal{U}$ is a complete category. In particular, we denote $X \times^k Y := k(X \times Y)$ as the product of two k -spaces X, Y .

To show it's also cocomplete, consider the following:

Proposition 1.6. *Quotient of a k -space is still a k -space; arbitrary disjoint union of k -spaces is a k -space.*

Proof. Let X be a k -space, and $f : X \rightarrow Y$ be a quotient map. For any k -closed subset $A \subseteq Y$, consider arbitrary continuous map $g : K \rightarrow X$ with K compact. Given the composition $f \circ g : K \rightarrow X \rightarrow Y$, $(f \circ g)^{-1}(A) = g^{-1}(f^{-1}(A)) \subseteq K$ is closed, hence $f^{-1}(A) \subseteq X$ is k -closed, which is closed as X is a k -space. Then, using the quotient space property, $A = f(f^{-1}(A)) \subseteq Y$ is closed, concluding that Y is also a k -space.

Now, given $\{X_i\}_i$ a family of k -spaces, consider $Z := \coprod_i X_i$, with $\iota_i : X_i \rightarrow Z$ the canonical inclusion. Given any k -closed subset $A \subseteq Z$, similar logic from previous proofs show that $A \cap X_i = \iota_i^{-1}(A) \subseteq X_i$ are all k -closed, hence closed (as X_i 's are k -spaces). Which, this verifies $A \subseteq Z$ is closed under the disjoint union topology, showing Z is also a k -space. $\square$

This guarantees the coproduct and coequalizers of k -spaces (and maps between them) in $\mathcal{U}$ are still k -spaces, which these colimits exist in $k\mathcal{U}$, showing the cocompleteness.

Now, move on to another criterion for compactly generated spaces:

Definition 1.7. A space X is *k-Hausdorff* if the diagonal $\Delta X \subset k(X \times X)$ is closed.

Denote $h\mathcal{U}$ as the *Category of k-Hausdorff spaces*.

There's a lot more details for this, I'll try and fill it in afterward==.

There is another functor $h : \mathcal{U} \rightarrow h\mathcal{U}$ which generates a *k-Hausdorff* space out of any topological spaces, and with inclusion functor $\mathcal{J} : h\mathcal{U} \rightarrow \mathcal{U}$, they form an adjoint pair $(h, \mathcal{J})$. So, h as a left adjoint preserves colimits.

Finally, for compactly generated spaces, it's defined as a *k-Hausdorff k-space*. In this space, limits and colimits exist, but need to be constructed in $\mathcal{U}$, then apply the functor h or k to yield the desired results.

Another equivalent way to phrase *k-Hausdorffness* when given a *k-space*, is the *weak Hausdorffness*: Given any continuous map $g : K \rightarrow X$ with K compact, $g(K) \subseteq X$ is a closed subset.

1.2 Spaces of Maps

This section, we'd like to realize $Y^X := \text{Hom}_{\text{Top}}(X, Y)$ (the set of all continuous maps $X \rightarrow Y$) as a topological space. Here, we consider the *compact open topology*.

First, given the any map $f : K \times X \rightarrow Y$, there is a corresponding set function $\tilde{f} : K \rightarrow Y^X$ by $\tilde{f}(k) = f(k, _) : X \rightarrow Y$.

Definition 1.8. The *compact open topology* on $\text{Hom}_{\text{Top}}(X, Y)$, has $S \subseteq Y^X$ being open iff all compact space K , continuous map $f : K \times X \rightarrow Y$, the set

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, _) \in S\}$$

is open in K .

Obviously, $\emptyset, Y^X$ are both open; if $\{S_i \subseteq Y^X\}_i$ are open, then $\tilde{f}^{-1}(\bigcup_i S_i) = \bigcup_i \tilde{f}^{-1}(S_i)$ is open in K , so $\bigcup_i S_i$ is open; finally, if $S, S' \subseteq Y^X$ are open, then $\tilde{f}^{-1}(S \cap S') = \tilde{f}^{-1}(S) \cap \tilde{f}^{-1}(S')$ is open in K , so $S \cap S'$ is also open. This defines a topology.

Here is some nice properties of composition of maps:

Proposition 1.9. Given $g : X' \rightarrow X$, the precomposition $g^* := _ \circ g : Y^X \rightarrow Y^{X'}$ is continuous. Dually, given $h : Y \rightarrow Y'$, the postcomposition $h_* := h \circ _ : Y^X \rightarrow Y'^X$ is continuous.

Proof. In the first scenario, given arbitrary open subset $S' \subseteq Y^{X'}$, the preimage $S := (g^*)^{-1}(S') = \{\ell \in Y^X | \ell \circ g \in S'\}$. Fix arbitrary continuous map $f : K \times X \rightarrow Y$ with K compact, consider $\tilde{f}^{-1}(S) \subseteq K$, we get:

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, _) \in S\} = \{k \in K | f(k, _) \circ g \in S'\}$$

Similarly, consider $f' := f \circ (\text{id}_K \times g) : K \times X' \rightarrow Y$ (by $f'(k, x') = f(k, g(x'))$), the set $\tilde{f}'^{-1}(S') \subseteq K$ is open by assumption, which it has the following description:

$$\tilde{f}'^{-1}(S') = \{k \in K | f'(k, _) \in S'\} = \{k \in K | f(k, _) \circ g \in S'\} = \tilde{f}^{-1}(S)$$

This shows the openness of $S \subseteq Y^X$.

For the second scenario, given arbitrary open subset $S' \subseteq Y'^X$, the preimage $S := h_*^{-1}(S') = \{\ell \in Y^X | h \circ \ell \in S'\}$. Fix arbitrary continuous map $f : K \times X \rightarrow Y$ with K compact, notice the following description:

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, _) \in S\} = \{k \in K | h \circ f(k, _) \in S'\}$$

Which, notice $h \circ f : K \times X \rightarrow Y \rightarrow Y'$ satisfies $(h \circ f)^{-1}(S') = \{k \in K | h \circ f(k, _) \in S'\} = \tilde{f}^{-1}(S)$, and by assumption this set is open. So, $S \subseteq Y^X$ is open, finishing the claim. $\square$

As a result, the Hom functor $\text{Hom}_{\text{Top}} : \text{Top}^{\text{op}} \times \text{Top} \rightarrow \text{Top}$ is a bifunctor.

If restrict to the compactly generated spaces, there is a homeomorphism

$$\text{Hom}_{\text{Top}}(X \times^k Y, Z) \cong \text{Hom}_{\text{Top}}(X, \text{Hom}_{\text{Top}}(Y, Z))$$

Yeah, more details to fill in.

1.3 Cofibration

Intuitively, these are the maps where homotopy can be extended to the codomain:

Definition 1.10. A map $\iota : A \rightarrow X$ is a *cofibration*, if it satisfies the *Homotopy ExtensionProperty* (HEP). This means, given any map $f : X \rightarrow Y$, homotopy $H : A \times I \rightarrow Y$, zeroth inclusions $i_0 : A \rightarrow A \times I$ and $i_0 : X \rightarrow X \times I$ (by $x \mapsto (x, 0)$), and product map $\iota \times \text{id}_I : A \times I \rightarrow X \times I$, the homotopy extends to $\tilde{H} : X \times I \rightarrow Y$, with the following commutes:

$$\begin{array}{ccc}
 A & \xrightarrow{i_0} & A \times I \\
 \downarrow \iota & \nearrow H & \downarrow \iota \times \text{id}_I \\
 & Y & \\
 \uparrow f & \nwarrow \exists \tilde{H} & \\
 X & \xrightarrow{i_0} & X \times I
 \end{array}$$

Intuitively, it gives a description of how to map X into Y (also how to map A into Y), generates a homotopy H on A only, and the property guarantees it to extend to a homotopy on X .

It's preserved under fibre product:

Lemma 1.11. If $\iota : A \rightarrow X$ is a cofibration, $g : A \rightarrow B$ any map, then the induced pushout $\iota' : B \rightarrow B \sqcup_A X$ is a cofibration:

$$\begin{array}{ccc}
 A & \xrightarrow{\iota} & X \\
 g \downarrow & & \downarrow g' \\
 B & \xrightarrow{\iota'} & B \sqcup_A X
 \end{array}$$

Proof. Before starting, notice that product preserves adjunction/fibre coproduct, because $(B \sqcup_A X) \times I \cong (B \times I) \sqcup (A \times I)(X \times I)$ (i.e. the first space can be viewed as the disjoint union of the product, then glue using identical methods at each $t \in I$).

Then, given map $f : B \sqcup_A X \rightarrow Y$, homotopy $H : B \times I \rightarrow Y$, and the fibre coproduct maps, we get:

$$\begin{array}{ccccc}
 A & & \xrightarrow{i_0} & & A \times I \\
 \downarrow \iota & \searrow g & & \swarrow g \times \text{id}_I & \downarrow \iota \times \text{id}_I \\
 & B & \xrightarrow{i_0} & B \times I & \\
 & \downarrow \iota' & \nearrow H & \downarrow \iota' \times \text{id}_I & \\
 & & Y & & \\
 & \nearrow f & \nwarrow \exists \tilde{H} & \nearrow g' \times \text{id}_I & \\
 & B \sqcup_A X & \xrightarrow{i_0} & (B \sqcup_A X) \times I & \\
 \uparrow g' & & & & \downarrow \\
 X & & \xrightarrow{i_0} & & X \times I
 \end{array}$$

The existence of $\tilde{H}$ is given by HEP of $\iota : A \rightarrow X$. This induces the following diagram:

$$\begin{array}{ccc}
A \times I & \xrightarrow{\iota \times \text{id}_I} & X \times I \\
\downarrow g \times \text{id}_I & & \downarrow g' \times \text{id}_I \\
B \times I & \xrightarrow{\iota' \times \text{id}_I} & (B \sqcup_A X) \times I \\
& \searrow H & \swarrow \exists! \bar{H} \\
& & Y
\end{array}$$

(A curved arrow $\tilde{H}$ also goes from $X \times I$ to Y)

where $\bar{H}$ is induced by the fibre coproduct property. Plug in to the previous diagram, we yield the desired homotopy extension for the inner square. $\square$

If consider the “special case” $Y = M_\iota$, the mapping cylinder of $\iota : A \rightarrow X$ (which is constructed as $M_\iota := (A \times I \sqcup X) / \sim$, where $(a, 0) \sim f(a)$). Notice it generates the following fibre coproduct diagram:

$$\begin{array}{ccc}
A & \xrightarrow{i_0} & A \times I \\
\downarrow \iota & & \downarrow H \\
X & \xrightarrow{i'_0} & M_\iota
\end{array}$$

So, the HEP and fibre coproduct property guarantees the following diagram:

$$\begin{array}{ccccc}
A & \xrightarrow{i_0} & A \times I & & \\
\downarrow \iota & & \downarrow H & \nearrow \tilde{H} & \\
& & M_\iota & \nwarrow \exists! j & \\
& \nearrow i'_0 & & \nwarrow \exists! j & \\
X & \xrightarrow{i_0} & X \times I & & \\
& & \downarrow \iota \times \text{id}_I & &
\end{array}$$

Here, $\tilde{H}$ is the homotopy extension, j is the factorization by fibre coproduct. And, this enforces $\tilde{H} \circ j = \text{id}_{M_\iota}$, showing j is injective. If restrict to $t = 1$, then $j : A \times \{1\} \rightarrow X \times \{1\}$ is precisely the map ι , showing it's injective (in fact it's an embedding, because we also have a retract $\tilde{H} : X \times \{1\} \rightarrow A \times \{1\}$).

From other sources I can find, ι unfortunately doesn't have closed image in general, yet for the case of A, X being compactly generated (for instance, CW Complexes) I believe this is true. (Maybe by k -closed property).

So, we see $\iota : A \rightarrow X$ is a cofibration (for compactly generated spaces) directly implies ι is an embedding with closed image (or A is a closed subspace of X); and, the mapping cylinder is precisely $M_\iota = X \times \{0\} \cup A \times I$ in $X \times I$, the extended homotopy guarantees it's a retract of $X \times I$.

Conversely, given $A \subseteq X$ a closed subspace, while its mapping cylinder $M_\iota \subseteq X \times I$ is a retract of $X \times I$. Then, use the retract as the extended homotopy (together with the fact that it's a fibre coproduct, hence the “initial extension diagram” whenever seeking the homotopy extension for other Y), we get $A \hookrightarrow X$ is a cofibration.

So, for compactly generated spaces, it's equivalent to say: a cofibration is a closed embedding $A \subseteq X$, such that the mapping cylinder $X \times \{0\} \cup A \times I$ is a retract of $X \times I$.

To talk about homotopy equivalences of cofibers (which are not necessarily cofibrations, but just maps $A \rightarrow X$ when fixing A). Given, $i : A \rightarrow X$, $j : A \rightarrow Y$ two cofibers, a map $f : X \rightarrow Y$ is a morphism, if the following diagram commutes:

$$\begin{array}{ccc} & A & \\ i \swarrow & & \searrow j \\ X & \xrightarrow{f} & Y \end{array}$$

Definition 1.12. Two such morphisms $f, f' : X \rightarrow Y$ are homotopic in this category (under A), if they're homotopic relative to A , i.e. there exists homotopy $H : X \times I \rightarrow Y$ of f, f' , where all $a \in A, t \in I$ satisfies $H(i(a), t) = j(a)$ (namely A is fixed under this homotopy). This is called *cofiber homotopy*.

The homotopy equivalence built on cofiber homotopies is called *cofiber homotopy equivalence*.

This notion of homotopy equivalence is much stronger than original homotopy. Yet, when restricting to cofibrations, then it doesn't make a difference:

Proposition 1.13. *Given $i : A \rightarrow X$, $j : A \rightarrow Y$ two cofibrations, if morphism $f : X \rightarrow Y$ is a homotopy equivalence, then it's a cofiber homotopy equivalence.*

Proof. It suffices to find $g : Y \rightarrow X$ that's a homotopy inverse, with $g \circ f \simeq \text{id}_X$ relative to A (because then apply the same argument on $g : Y \rightarrow X$, there is a homotopy inverse $f' : X \rightarrow Y$, such that $f' \circ g \simeq \text{id}_Y$ relative to A . As a result, $f = \text{id}_Y \circ f \simeq f' \circ g \circ f \simeq f' \circ \text{id}_X = f'$ relative to A , hence $f \circ g \simeq f' \circ g \simeq \text{id}_Y$ relative to A , showing it's a cofiber homotopy equivalence).

First, by hypothesis, there is $g'' : Y \rightarrow X$ a homotopy inverse of f . (Not done)

□

1.4 Fibration

As a dual notion to cofibration (which “extends” homotopy to the target space), fibration is the one that “lifts” the homotopy to the source space.

Definition 1.14. A surjective map $p : E \twoheadrightarrow B$ is a *fibration*, if it satisfies *covering homotopy property* (CHP in May’s book), or *homotopy lifting property*. Meaning, given any $f : Y \rightarrow E$, and homotopy $H : Y \times I \rightarrow B$ of $p \circ f : Y \rightarrow E \twoheadrightarrow B$, there exists a homotopy $\tilde{H} : Y \times I \rightarrow E$ of f , with the following diagram comutes:

$$\begin{array}{ccc} Y & \xrightarrow{\forall f} & E \\ i_0 \downarrow & \exists \tilde{H} \nearrow & \downarrow p \\ Y \times I & \xrightarrow{\forall H} & B \end{array}$$

A typical example is the covering map, where homotopies can be lifted uniquely once a basepoint of the covering is chosen. So, fibration can be viewed as a generalization of covering space, where homotopy liftings exist.

Using the k -ified compact-open topology of the function space (in particular, the homeomorphism $\text{Hom}_{\text{Top}}(K \times X, Y) \cong \text{Hom}_{\text{Top}}(K, \text{Hom}_{\text{Top}}(X, Y))$, and $Y^X := \text{Hom}_{\text{Top}}(X, Y)$), the above can be reformulated as the following (which is structure wise more similar to the dual of the cofibration diagram):

$$\begin{array}{ccc} E & \xleftarrow{p_0} & E^I \\ \downarrow p & \exists \tilde{H} \nearrow & \downarrow p \circ - \\ Y & & Y \\ \downarrow \forall f & & \downarrow \forall H \\ B & \xleftarrow{p_0} & B^I \end{array}$$

Then, by dualizing the proof for cofibration, we yield the following result:

Lemma 1.15. *Pullback of fibration is fibration: Given fibration $p : E \twoheadrightarrow B$, and any map $g : B' \rightarrow B$, the base change $p' : E \times_B B' \rightarrow B'$ is a fibration:*

$$\begin{array}{ccc} E \times_B B' & \xrightarrow{g'} & E \\ p' \downarrow & & \downarrow p \\ B' & \xrightarrow{g} & B \end{array}$$

Similar to the existence of mapping cylinder M_ι as the “initial test diagram” for cofibration (as it’s the fibre coproduct of cofibration $A \rightarrow X$, and inclusion $A \rightarrow A \times I$), it suggests the existence of “final test diagram” for fibration, when choosing $Y = N_p := E \times_B B^I$ (the fibre product of fibration $E \rightarrow B$, and evaluation by zero map $B^I \rightarrow B$). The fibre product property (together with the test diagram of fibration) guarantees N_p is the final diagram (namely, all other choices of Y factors uniquely through N_p).

A precise description of N_p is as follow:

$$N_p = \{(e, \beta) | \beta(0) = p(e)\} \subseteq E \times B^I$$

Plugin to the diagram, we observe the following using the fibration and fibre product property:

$$\begin{array}{ccc}
 E & \xleftarrow{p_0} & E^I \\
 \downarrow p & \nearrow p_1 & \downarrow p \circ - \\
 & N_p & \\
 & \searrow p_2 & \downarrow p \circ - \\
 B & \xleftarrow{p_0} & B^I
 \end{array}$$

$\exists! k$ (dashed arrow from E to N_p)
 $\exists s$ (dashed arrow from N_p to E^I)

Here, k is induced by fibre coproduct, s is induced by fibration property. Universality of N_p (as fibre product) guarantees that $k \circ s = id_{N_p}$, another name for these lifting s is called *path lifting function* (as it lifts a pair $(e, \beta) \in N_p$ with $\beta(0) = p(e)$, to a path $\tilde{\beta} \in E^I$, that satisfies $\tilde{\beta}(0) = e$, and $p \circ \tilde{\beta} = \beta \in B^I$).

A special case - when $p : E \rightarrow B$ is a covering, then s is in fact unique, due to the unique path lifting property (whenever fixing a path and a basepoint).

For common types of fibrations, the case for covering space suggests the notion of fibre bundle could work: Given covering $p : E \rightarrow B$, every point $x \in B$ has an open neighborhood $U_x \subseteq B$, such that $p^{-1}(U_x) \cong U_x \times p^{-1}(x)$ (where $p^{-1}(x)$ is a discrete subspace), and all fibres are homeomorphic to each other (i.e. same cardinality when the fibre is discrete).

If restricts the bundle to have *numerable open covers* (namely, an open cover $\{U_i\}_i$, such that each point $x \in B$ is contained in finitely many U_i 's, and each U_i is equipped with a continuous function $\lambda_i : B \rightarrow I$ such that $\lambda_i^{-1}(0, 1] = U_i$). Then, the following theorem holds:

Theorem 1.16. *Given a map $p : E \rightarrow B$ and $\{U_i\}_i$ a numerable open cover of B . Then, p is a fibration iff the restrictions $p|_{U_i} : p^{-1}(U_i) \rightarrow U_i$ is a fibration for every U_i .*

This guarantees all fibre bundles with numerable trivialization cover to be a fibration.

Similar to cofibre homotopy, we define fibre homotopy as follow:

Definition 1.17. Given two fibres $p : D \rightarrow B$ and $p' : E \rightarrow B$ (Note: they're not necessarily fibrations), a map $f : D \rightarrow E$ is a morphism if the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{f} & E \\
 \searrow p & & \swarrow p' \\
 & B &
 \end{array}$$

And, given two morphisms $f, f' : D \rightarrow E$, they're *fibre homotopic* if there is a homotopy $H : D \times I \rightarrow E$, such that $p = p' \circ H : D \rightarrow B$ (namely, at all time $t \in I$, the map preserves p).

By dualizing the notion from cofibre, we again have the following criterion of fibre homotopy equivalence:

Proposition 1.18. *Let $p : D \rightarrow B$, $q : E \rightarrow B$ be two fibrations. If $f : D \rightarrow E$ be a homotopy equivalence morphism (i.e. $q \circ f = p$), then it's a fibre homotopy equivalence.*

To talk about the relations between the fibres (for instance, for fibre bundle all fibres are homeomorphic), the use of paths (as homotopy between points) can also be lifted given fibration, and can be used to study such relation.

Let $p : E \rightarrow B$ be a fibration, $\beta : I \rightarrow B$ be a path with $\beta(0) = b$ and $\beta(1) = b'$, and $F_x := p^{-1}(x) \subseteq E$ as the fibre (as subspace under E). Then, one can consider the following diagram, using the lifting property:

$$\begin{array}{ccccc}
 F_b \times \{0\} \cong F_b & \xrightarrow{\iota_b} & E & & \\
 \downarrow i_0 & \nearrow \exists \tilde{\beta} & \downarrow p & & \\
 F_b \times I & \xrightarrow{\pi_2} I & \xrightarrow{\beta} B & &
 \end{array}$$

Which, specify to $t = 1$ for $\tilde{\beta}$, we realize the following:

$$p \circ \tilde{\beta}_1(F_b) = \beta \circ \pi_2(F_b \times \{1\}) = \beta(1) = b' \implies \tilde{\beta}_1(F_b) \subseteq F_{b'}$$

So, it induces a map $\tilde{\beta}_1 : F_b \rightarrow F_{b'}$ (more generally, it induces a continuous family of maps $\tilde{\beta}_t : F_b \rightarrow F_{\beta(t)}$).

Proposition 1.19. *Suppose paths $\beta, \beta' : I \rightarrow B$ are homotopic, then the induced maps $\tilde{\beta}_1, \tilde{\beta}'_1 : F_b \rightarrow F_{b'}$ are also homotopic.*

Proof. Let $H : I \times I \rightarrow B$ be the path homotopy of $\beta \simeq \beta'$, so that $H(s, 0) = \beta(s)$, $H(s, 1) = \beta'(s)$, $H(0, t) = b$, and $H(1, t) = b'$. If consider the subspace of $I \times I$, say $J := (I \times \{0\}) \cup (I \times \{1\}) \cup (\{0\} \times I)$ (in the homotopy H , first side represents β , second side represents β' , and the third side represents the side that's constant as b), notice there is an automorphism on $I \times I$, so that J gets mapped to $I \times \{0\}$, so the lifting property can be used whenever a map is defined on $J \subset I \times I$.

Now, define the map $f : F_b \times J \rightarrow E$, so that on $I \times \{0\}$ it's $\tilde{\beta} : F_b \times I \cong F_b \times I \times \{0\} \rightarrow E$, on $I \times \{1\}$ it's $\tilde{\beta}' : F_b \times I \cong F_b \times I \times \{1\} \rightarrow E$, and on $\{0\} \times I$ it's the map $\iota_b \circ p_1 : F_b \times I \cong F_b \times \{0\} \times I \rightarrow F_b \rightarrow E$ (more precisely, we only consider the value in F_b , and include it into E). By the defining property of each $\tilde{\beta}, \tilde{\beta}'$, these three maps glue together, hence f is well-defined.

Then, consider the inclusion $i : J \hookrightarrow I \times I$, we form the following diagram using the lifting property:

$$\begin{array}{ccccc}
 F_b \times J & \xrightarrow{f} & E & & \\
 \downarrow \text{id}_{F_b} \times i & \nearrow \exists \tilde{H} & \downarrow p & & \\
 F_b \times I \times I & \xrightarrow{\pi_{2,3}} I \times I & \xrightarrow{H} B & &
 \end{array}$$

Which, if restrict to $F_b \times I \times \{0\}$, $\tilde{H}$ restricts to $\tilde{\beta}$ by our choice of f ; similarly, restricting to $F_b \times I \times \{1\}$, $\tilde{H}$ restricts to $\tilde{\beta}'$, showing the two homotopic. So, specify to $t = 1$, we get $\tilde{\beta}_1 \simeq \tilde{\beta}'_1$, showing the map $[\tilde{\beta}_1] : F_b \rightarrow F_{b'}$ is a well-defined map between fibres, and dependent only on homotopy classes of paths $[\beta] : I \rightarrow B$, connecting b to b' . $\square$

Corollary 1.20. *Given a fibration, any two points in the same path component have homotopy equivalent fibres.*

Proof. This is because if $\beta, \gamma : I \rightarrow B$ are two paths such that $\beta(1) = \gamma(0)$, then if $b := \beta(0), b' := \beta(1) = \gamma(0), b'' := \gamma(1)$, then $\tilde{\beta}_1 : F_b \rightarrow F_{b'}$ and $\tilde{\gamma}_1 : F_{b'} \rightarrow F_{b''}$ composes to be the same as $(\gamma \cdot \beta)_1 : F_b \rightarrow F_{b''}$ (it's induced by the concatenation of maps). So, $[\gamma \cdot \beta]_1 = [\tilde{\gamma}_1] \circ [\tilde{\beta}_1]$ when passing to homotopy classes of maps.

Then, since a constant path c_b includes identity $[\tilde{c}_b] : F_b \rightarrow F_b$, we see $[\beta^{-1}]_1 \circ [\tilde{\beta}_1] = [\beta^{-1} \cdot \beta]_1 = [\tilde{c}_b] = [\text{id}_{F_b}]$, showing $[\tilde{\beta}_1]$ is a homotopy equivalence. $\square$

1.5 Based Homotopy, Suspension, and Loop Spaces

For stable homotopy theory, I believe we're mainly working with spaces that fix a basepoint, and study the based homotopy between based maps. Which, let $\mathcal{C}$ denotes the category of based spaces, with $F(X, Y)$ denotes the morphisms (all the based maps $X \rightarrow Y$, meaning it preserves basepoint). Notice $F(X, Y) \subseteq Y^X$ has a natural subspace topology, and it has a natural basepoint, namely the *constant basepoint map* c_{y_0} that sends everything in X to the basepoint $y_0 \in Y$.

Consider the following definition, which serve as a universal object in this category:

Definition 1.21. Let $X, Y \in \mathcal{C}$ equipped with basepoints $x_0 \in X, y_0 \in Y$. Let the *wedge sum* $X \vee Y := X \sqcup Y \setminus \{x_0, y_0\}$. Which, it naturally embeds into $X \times Y$ as a based space with basepoint (x_0, y_0) :

$$X \vee Y \rightarrow X \times Y, \quad x \mapsto (x, y_0), \quad y \mapsto (x_0, y)$$

Now, define the *smash product* $X \wedge Y := X \times Y / X \vee Y$, equipped with basepoint $x_0 \wedge y_0$ (the class of (x_0, y_0) under the quotient).

I want to complain a bit, why does wedge sum use vee instead of wedge, while smash product uses wedge...I'm always typing it in the wrong order.

Here, we assume $0 \in I, 1 \in S^1 \hookrightarrow \mathbb{C}$ are the corresponding basepoints of the two spaces. The other three important objects are as follow:

Definition 1.22. Given a based space X . Define the following:

- The *reduced cone* $CX := X \wedge I$.
- The *reduced suspension* $\Sigma X := X \wedge S^1$.
- The *loop space* $\Omega X := F(S^1, X)$.

For the latter two objects, it in fact has the following adjunction property:

Proposition 1.23. *Given based space X, Y , there is a natural homeomorphism $F(\Sigma X, Y) \cong F(X, \Omega Y)$.*

Proof. To establish a natural bijection is easy, for the topology I think I need to blackbox a bit...

For each based map $f : X \wedge S^1 \rightarrow Y$, it pullsback to the product map $f' : X \times S^1 \rightarrow Y$, where $f'(x, s) = f(x \wedge s)$. Using the adjunction property $\text{Hom}_{\text{Top}}(X \times S^1, Y) \cong \text{Hom}_{\text{Top}}(X, Y^{S^1})$, this induces $\tilde{f} : X \rightarrow Y^{S^1}$, where $\tilde{f}(x) = \tilde{f}_x : S^1 \rightarrow Y$ by $\tilde{f}_x(s) = f'(x, s)$.

Notice each $\tilde{f}_x$ is a based map, as $\tilde{f}_x(1) = f'(x, 1) = f(x \wedge 1) = y_0$ (as $x \wedge 1 = x_0 \wedge 1$ under smash product, and f is based). Also, notice $\tilde{f}_{x_0}$ is the based constant map, as $\tilde{f}_{x_0}(s) = f'(x_0, s) = f(x_0 \wedge s) = y_0$. So, $\tilde{f} : X \rightarrow F(S^1, Y) = \Omega Y$ in fact, and it's a based map.

This induces $F(\Sigma X, Y) \rightarrow F(X, \Omega Y)$, by $f \mapsto \tilde{f}$.

Conversely, for each based map $\tilde{g} : X \rightarrow F(S^1, Y)$, viewing $F(S^1, Y) \subseteq Y^{S^1}$ as a subspace, the adjunction on regular maps induce $g' : X \times S^1 \rightarrow Y$, by $g'(x, s) = \tilde{g}_x(s)$.

Since each $\tilde{g}_x$ is a based map, we have $g'(x, 1) = \tilde{g}_x(1) = y_0$; also, since $\tilde{g}$ itself is a based map, $\tilde{g}_{x_0}$ is a constant based map, hence $\tilde{g}_{x_0}(s) = y_0$ regardless of $s \in S^1$. Hence, all $(x, s) \in X \vee S^1$ are mapped to $y_0 \in Y$ under g' , showing g' factors uniquely to $g : X \wedge S^1 \rightarrow Y$, with $g(x_0 \wedge 1) = y_0$, hence a based map.

This induces $F(X, \Omega Y) \leftarrow F(\Sigma X, Y)$, by $\tilde{g} \mapsto g$.

Finally, it's a routine check that the two are actually mutual inverses, so I wouldn't note it down here (as I've done on paper). $\square$

If denote $[X, Y]$ as all the homotopy classes of based maps $X \rightarrow Y$ (which are the set of morphisms in $h\mathcal{C}$, the *homotopy category of based spaces*), we have the following corollary:

Corollary 1.24. *In $h\mathcal{C}$, the adjunction still holds, namely $[\Sigma X, Y] \cong [X, \Omega Y]$.*

Proof. Given two based maps $f, g \in F(X, Y)$, we have $f \simeq g$ iff there exists $H : X \times I \rightarrow Y$, such that $H(x, 0) = f(x)$, $H(x, 1) = g(x)$, and $H(x_0, t) = y_0$ (for based homotopy, it's required to fix the basepoint at all time).

Notice the final condition insists each $H_t : X \rightarrow Y$ is a based map, using the adjunction property of maps, the second statement is equivalent to a map $\tilde{H} : I \rightarrow F(X, Y) \subseteq \text{Hom}_{\text{Top}}(X, Y)$, showing $f \simeq g$ in $F(X, Y)$ iff they're connected by paths.

So, the natural homeomorphism $F(\Sigma X, Y) \cong F(X, \Omega Y)$ descends to a homeomorphism $[\Sigma X, Y] \cong [X, \Omega Y]$, as homotopy relation is equivalent to path connectedness on the space of based maps. $\square$

Notice as set, $[X, \Omega Y]$ has a group structure via *concatination*, as given $f, g \in F(X, \Omega Y)$, each $f(x), g(x) : S^1 \rightarrow Y$ is a based map, so concatination $g(x) \cdot f(x) : S^1 \rightarrow Y$ still makes sense. If define $(g \cdot f)(x) := g(x) \cdot f(x)$ in $F(X, \Omega Y)$, then the homotopy between loops guarantee such concatination is associative in $[X, \Omega Y]$.

Which, the homeomorphism also induces a concatination structure on $[\Sigma X, Y]$, with precise description as follow for all $f, g \in [\Sigma X, Y]$:

$$(g \cdot f)(x \wedge t) = \begin{cases} f(x \wedge 2t) & 0 \leq t \leq \frac{1}{2} \\ g(x \wedge (2t - 1)) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

Here, we view $S^1 := I/\partial I$ when constructing the concatination. If defining $f^{-1}(x \wedge t) := f(x \wedge (1 - t))$, the same proof as fundamental group guarantees this class of function is the inverse for $[f] \in [\Sigma X, Y]$, forming it as a group.

As a preliminary remark for the next result, smash product is associative up to homeomorphism (intuitively, given $(X \wedge Y) \wedge Z$, when pulling back to $X \times Y \times Z$, the points projecting down to $(x_0 \wedge y_0) \wedge z_0$ is precisely $(X \vee Y) \vee Z$, and wedge sum is associative up to homeomorphism).

Proposition 1.25. *$[\Sigma^2 X, Y]$ is abelian.*

Proof. Given $\Sigma^2 X = X \wedge S^1 \wedge S^1$, the standard concatination uses the last entry (as demonstrated above). Yet, we also have concatination using the second entry also! So, let $\cdot_2$ denote the second entry concatination, $\cdot_3$ denote the third entry concatination, the following holds (if view horizontal as 2, vertical as 3):

$$\begin{pmatrix} g \\ f \end{pmatrix} = \begin{pmatrix} g & c \\ c & f \end{pmatrix} = \begin{pmatrix} g & f \end{pmatrix} = \begin{pmatrix} c & f \\ g & c \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}$$

Here, vertical vector means we're concatinating using the third entry, horizontal vector means concatinating using the second entry, and c denotes the constant map (i.e. identity) we're concatinating with.

This shows the commutativity. $\square$

Corollary 1.26. *All $n \geq 2$ has $[\Sigma^n X, Y]$ being abelian, by applying **Proposition 1.25** inductively.*

1.6 Based Homotopy of Fibres, Cofibres, and Adjunction (Σ, Ω)